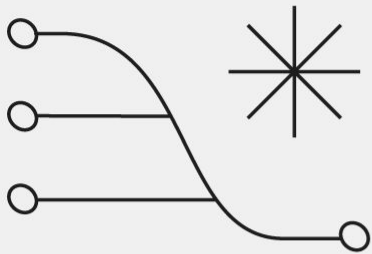




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Building AI Agents With Google ADK



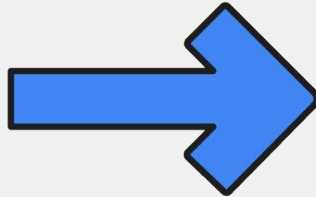
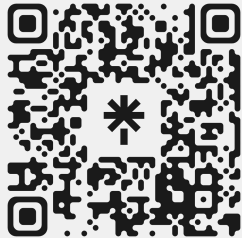
Build  with AI



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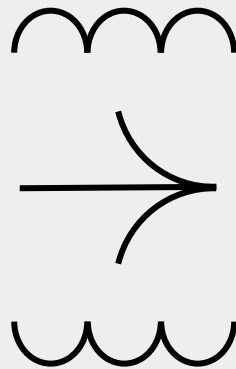
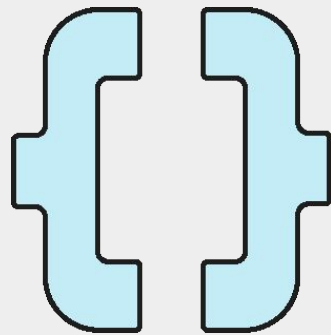
Muhammad Rebaal

AI Automation Engineer @ZerOwl



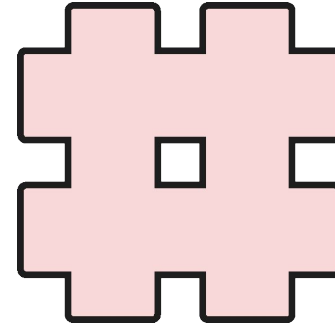
Community

We are the Google Developer Groups on Campus at the University of Karachi a community driven by curiosity and a passion for technology. Our mission is to empower students to learn cutting-edge skills, tackle real-world problems, and prepare for successful careers in tech. Whether you want to build your technical stack, connect with industry mentors, or improve your soft skills like problem-solving and communication, GDGoC-UoK is your platform to grow. Led by tech enthusiasts Muhammad Rebaal and Syeda Rida Fatima, we welcome all students to join our diverse and dynamic learning environment!



Agenda

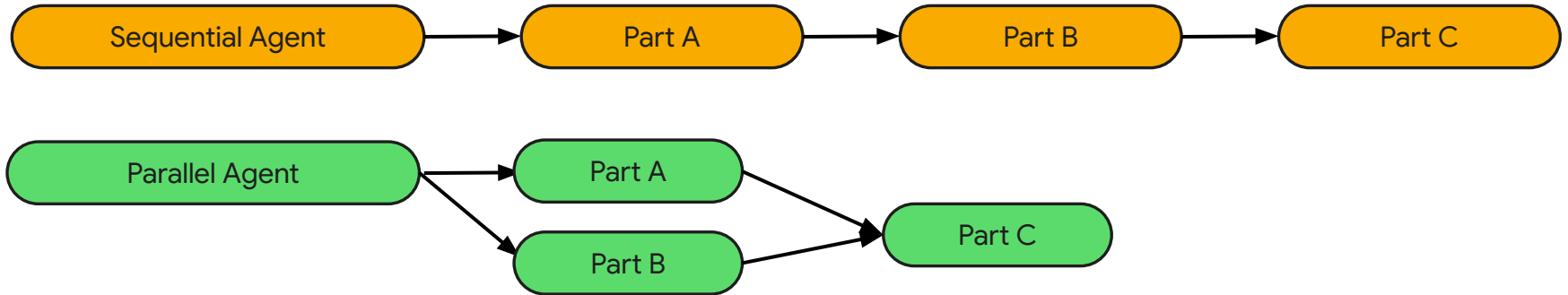
- 1 What is an AI Agent ?
- 2 Architecture of ADK
- 3 Context & State Management
- 4 Communication Protocols
- 5 Demo



What is an AI Agent ?

AI Agent is a system that uses an AI model (like a brain) to make decisions and take actions to achieve a specific goal.

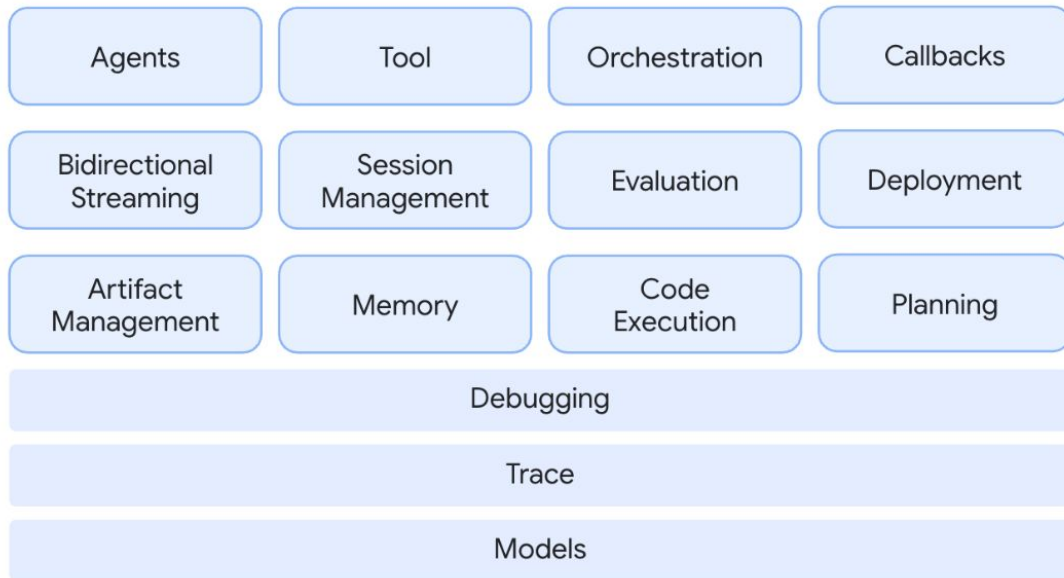
Types of Agent



Architecture Of ADK

Agent Development Kit

Client side SDK to define multi-agent applications for complex, real world scenarios



Context & State Management

Sessions

- The "short-term" memory of a single conversation thread. Ends when the task is done.

Memory Bank

- Long-term facts (preferences, allergies, history) that persist across months.

Artifacts

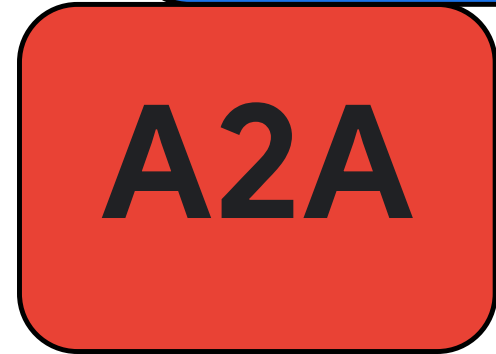
- Management of non-text files like PDFs, images, and spreadsheets generated by agents.

Communication Protocols

Communication Protocol: is an agreed-upon set of rules that allows two different entities (like computers, software programs, or AI models) to exchange information.

MCP: is an open standard recently introduced to standardize how AI models connect to external data sources and tools.

A2A: refers to the standards and rules that allow different, autonomous AI agents to communicate, negotiate, and collaborate with one another to complete complex tasks.



Demo

Thank You

Join Community

